India-Centric Traffic AI Project
Dataset Acquisition Plan

Practical shortlist: downloadable + actually useful datasets

1. What we actually need

The project should not depend on one giant dataset. We will combine Indian traffic datasets for vehicle detection, helmet detection and ANPR, then validate the complete pipeline on Indian traffic footage. The immediate Tier-1 pipeline is: vehicle/person detection → ByteTrack → rider association → helmet/triple-riding verification → plate detection → OCR → evidence.

2. Download first — highest priority

Dataset	Use in project	Status / access	Priority
UVH-26 (IISc AIM)	Indian CCTV vehicle detection; Indian traffic adaptation/evaluation	Public on Hugging Face; 26,646 1080p frames, ~1.8M boxes, 14 vehicle classes	★★★★★
Indian Traffic Helmet Violation (Hugging Face)	Helmet vs no-helmet training	Public; 42,559 images, ~126k boxes; CC BY 4.0	★★★★★
Indian Vehicle License Plate (Kaggle)	Indian plate detection + OCR training/evaluation	Public Kaggle download; includes Maharashtra highway video frames and XML plate/text annotations	★★★★★
IDD-D YOLOv11 (Kaggle)	Indian road detection; motorcycle, rider, person, auto, car, bus, truck, etc.	Public Kaggle download; 41,962 labeled images; CC BY-NC-SA 4.0	★★★★☆

3. Useful secondary datasets

Dataset	Why keep it	Recommendation
ITD — Indian Traffic Dataset v1.2	9,200+ annotated images, 280k+ objects, 25+ locations across 14 Indian states/UTs; strong Indian traffic benchmark	Request access; use when obtained
IITH Helmet datasets	Real surveillance videos from IIT Hyderabad / Hyderabad CCTV; useful for video-level helmet testing	Request/download if accessible
Indian number plate YOLO annotations (Kaggle)	Indian number-plate detector in YOLO format; CC0	Good backup for plate detector
IITM-HeTra	1,417 reviewed frames from Chennai traffic; heterogeneous Indian traffic	Useful validation/extra vehicle data
Indian Road Dataset (Atal Setu Mumbai)	4,000+ images/videos; Indian road and lane-marking context; MIT	Save for later infrastructure/lane module

4. Datasets we should NOT download right now

Do not download everything just because it is India-related. DataCluster's larger vehicle/plate collections may contain useful samples, but full access can require a license/contact with the provider. They are optional, not dependencies. The MOTOR two-wheeler behavior dataset is also large (~106 GB) and is better saved for a later traffic-behavior phase.

5. Exact project dataset stack

- Vehicle detector: start from your existing best.pt; use UVH-26 + IDD-D for Indian fine-tuning/evaluation if needed.
- Helmet detector: use the 42,559-image Indian Traffic Helmet Violation dataset.
- Plate detector: use the Indian Vehicle License Plate dataset; optionally add the CC0 YOLO plate dataset.
- OCR: use Indian plate crops and repeated OCR results across frames; keep a separate held-out test set.
- Tracking: ByteTrack does not require a separate tracking-training dataset for the initial implementation.
- Final system validation: use real Indian traffic/intersection video, ideally a fixed CCTV-style camera.

6. Recommended folder layout

```
AI-Traffic-System/
├── datasets/
│   ├── uvh26/
│   ├── idd/
│   ├── helmet_india/
│   ├── plates_india/
│   └── validation_video/
├── models/
│   ├── traffic.pt
│   ├── helmet.pt
│   └── plate.pt
├── data/
│   ├── videos/
│   └── evidence/
└── database/
    └── violations.db
```

7. Links / access points

- **UVH-26 — IISc AIM / Hugging Face:** https://huggingface.co/datasets/iisc-aim/UVH-26
- **Indian Traffic Helmet Violation — Hugging Face:** https://huggingface.co/datasets/thundarstrom/traffic-helmet-violation
- **Indian Vehicle License Plate Dataset — Kaggle:** https://www.kaggle.com/datasets/saisirishan/indian-vehicle-dataset
- **IDD-D YOLOv11 — Kaggle:** https://www.kaggle.com/datasets/redzapdos123/indian-driving-dataset-detections-yolov11
- **ITD — Indian Traffic Dataset:** https://github.com/teg-iitr/ITD-Indian-traffic-dataset
- **IITH Helmet Dataset information:** https://sites.google.com/site/dineshsinghindian/iith_helmet_1-dataset
- **Indian plate YOLO annotations — Kaggle:** https://www.kaggle.com/datasets/deepakat002/indian-vehicle-number-plate-yolo-annotation

- **IITM-HeTra — Kaggle:** https://www.kaggle.com/datasets/deepak242424/iitmhetra
- **Indian Road Dataset — Kaggle:** https://www.kaggle.com/datasets/mansayy/indian-road-dataset

8. Important licensing / research notes

UVH-26 is listed with a CC BY 4.0 license. The Indian Traffic Helmet Violation dataset is CC BY 4.0. IDD-D is listed as CC BY-NC-SA 4.0 and is intended for academic/research use. The Indian Vehicle License Plate Kaggle dataset contains web-scraped and Maharashtra highway video-derived images; check its dataset terms before redistribution. ITD currently states that access is available for research use upon request.

9. Final recommendation

Download only these four first: UVH-26, Indian Traffic Helmet Violation, Indian Vehicle License Plate, and IDD-D. Keep ITD as a research-access request. Do not train anything until the files are checked for class names, annotation format, duplicates, image quality and licensing. Then we will build one clean India-centric dataset configuration rather than blindly merging incompatible labels.